

Question #1

In[1]:= $\{1, 3, 5\} + \{7, -2, -4\}$

Out[1]= $\{8, 1, 1\}$

Question #2

In[2]:= $\{1, 3, 5\} - \{7, -2, -4\}$

Out[2]= $\{-6, 5, 9\}$

Question #3

In[3]:= $6 \{1, 3, 5\}$

Out[3]= $\{6, 18, 30\}$

Question #4

In[4]:= $\text{Dot}[\{1, 3, 5\}, \{7, -2, -4\}]$

Out[4]= -19

Question #5

In[6]:= $\text{Norm}[\{1, 3, 5\}]$

Out[6]= $\sqrt{35}$

Question #6

In[10]:= $\text{Norm}[\{7, -2, -4\}]$

Out[10]= $\sqrt{69}$

Question #7

In[9]:= $\text{Solve}[\text{Dot}[\{1, 3, 5\}, \{7, -2, -4\}] == \text{Norm}[\{1, 3, 5\}] * \text{Norm}[\{7, -2, -4\}] * \text{Cos}[x], x]$

Out[9]= $\left\{ \left\{ x \rightarrow -\text{ArcCos}\left[-\frac{19}{\sqrt{2415}}\right] + 2\pi c_1 \text{ if } c_1 \in \mathbb{Z} \right\}, \left\{ x \rightarrow \text{ArcCos}\left[-\frac{19}{\sqrt{2415}}\right] + 2\pi c_1 \text{ if } c_1 \in \mathbb{Z} \right\} \right\}$

Question #8

In[11]:= $\text{Dot}[\{7, 3, -1, 4\}, \{5, 9, 2, 5\}]$

Out[11]= 80

Question #10 (a)

In[12]:= $\text{Cross}[\{2, 17, 5\}, \{-1, -4, -7\}]$

Out[12]= $\{-99, 9, 9\}$

Question #10 (b)

In[13]:= **Dot**[{2, 17, 5}, {-99, 9, 9}]

Out[13]=

0

In[14]:= **Dot**[{-1, -4, -7}, {-99, 9, 9}]

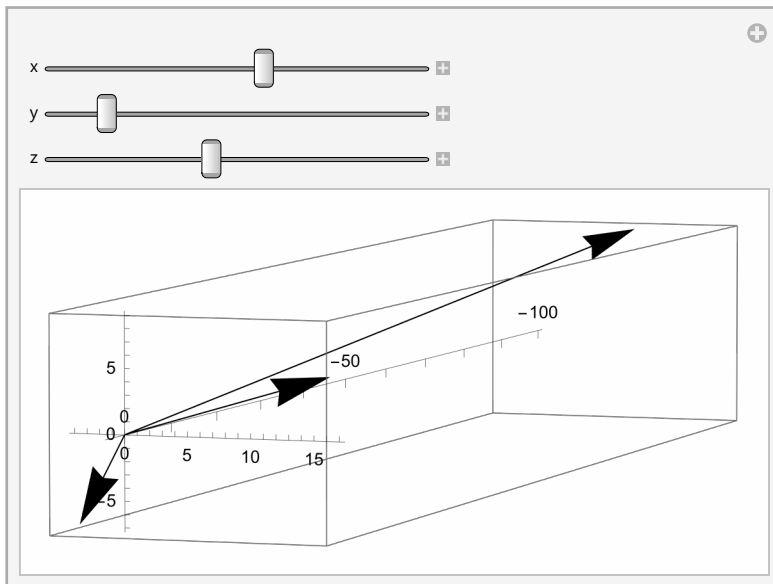
Out[14]=

0

Question #10 (c)

In[18]:= **Manipulate**[**Graphics3D**[{
 Arrow[{{0, 0, 0}, {2, 17, 5}}],
 Arrow[{{0, 0, 0}, {-1, -4, -7}}],
 Arrow[{{0, 0, 0}, {-99, 9, 9}}]
 }, **Axes** → **True**, **AxesOrigin** → {0, 0, 0}, **ViewPoint** → {x, y, z},
 {x, 1, 10}, {y, 1, 10}, {z, 1, 0}]

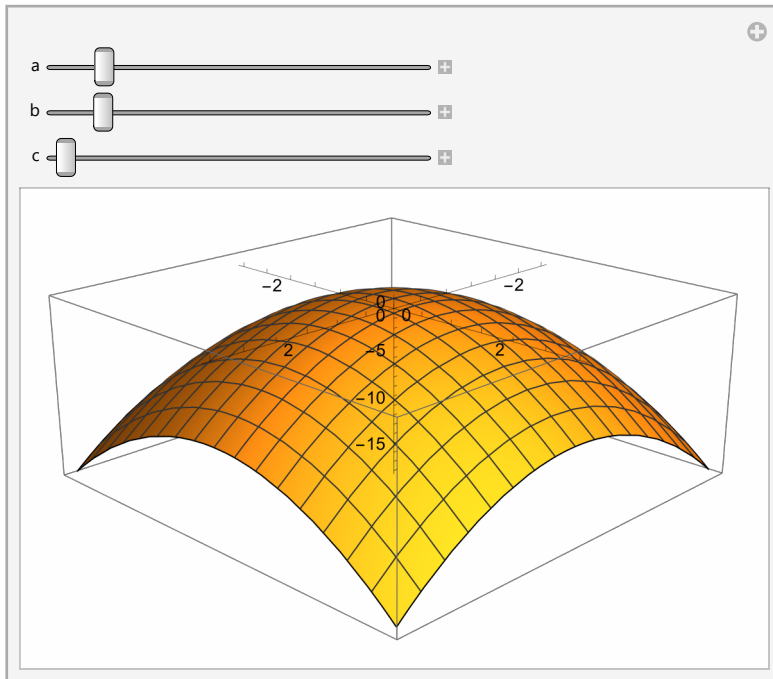
Out[18]=



Question #11 (a)

```
In[19]:= Manipulate[Plot3D[1 - x^2 - y^2, {x, -3, 3}, {y, -3, 3}, Axes → True,
  AxesOrigin → {0, 0, 0}, ViewPoint → {a, b, c}], {a, 1, 10}, {b, 1, 10}, {c, 1, 10}]
```

Out[19]=



Question #11 (b)

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In[23]:= Manipulate[Plot3D[Sqrt[1 - x^2 - y^2], {x, -1, 1}, {y, -1, 1}, Axes → True,  
  AxesOrigin → {0, 0, 0}, ViewPoint → {a, b, c}], {a, 1, 10}, {b, 1, 10}, {c, 1, 10}]
```

Out[23]=

